

Customer Shopping Behavior Analysis

By Sparsh Verma

Project Experience

- Movie Recommendation System – NLP on 4,500+ films (Python)
- Online Municipal Complaint Portal – Software Engineering
- Indian Entrepreneur Research – interviews on profitable businesses

Certifications

- Google Data Analytics (Coursera)
- Microsoft Excel: Beginner to Advanced (Udemy)



Skills

Research · Data Analysis (Python, Excel & Power BI) · Communication · Agility



Project Overview & Dataset



Dataset Summary

- **Simulated Retail Consumer Behavior Dataset**
- **3,900 rows**, 18 columns across demographics, purchase details, and shopping behavior
- 37 missing values in Review Rating – imputed using median per category
- Columns standardized to snake_case; promo_code_used dropped as redundant

Feature Engineering

- age_group via `pd.cut()`. (reflects real population distribution, avoids `qcut` bias)
- purchase_frequency_days derived from purchase data
- Cleaned DataFrame loaded into **PostgreSQL** for SQL analysis

SQL Analysis in PostgreSQL

Revenue & Spending

Revenue by gender;
High-spending discount users;
Subscribers vs. non-subscribers;
Revenue by age group

Products & Shipping

Top 5 products by rating;
Shipping type comparison;
Top 3 products per category via **window functions**
(`row_number()`)

Segmentation & Loyalty

New / Returning / Loyal via **CTE**;
Repeat buyers & subscriptions;
Bayes Theorem: $P(\text{Subscriber} \mid \text{Repeat Buyer}) = \mathbf{0.074}$



Power BI Dashboard

Customer Behavior Dashboard

Subscription Status

No

Yes

Gender

Female

Male

Categories

Accessories

Footwear

Clothing

Outerwear

Shipping Type

- ☐ 2-Day Shipping
- ☐ Express
- ☐ Free Shipping
- ☐ Next Day Air
- ☐ Standard
- ☐ Store Pickup

3.9K

No. of Customers

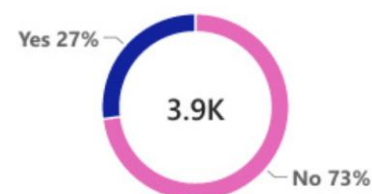
\$59.76

Average Purchase Amount

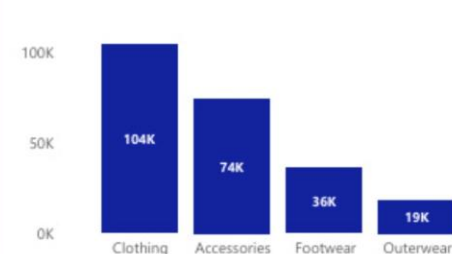
3.75

Average Review Rating

%age of Customers by Subscription Status



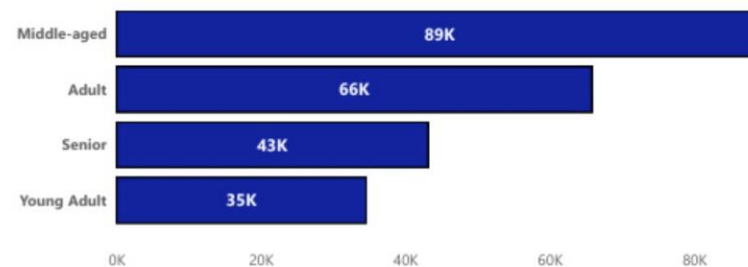
Revenue by Categories



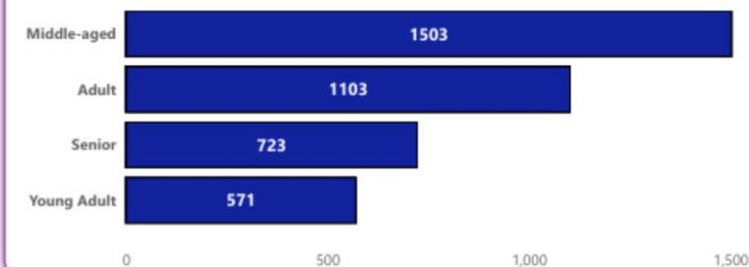
Sales by Categories



Revenue by Age Groups



Sales by Age Groups



Key Areas of Concern Identified Through Analysis

Gender Revenue Imbalance

Male: **\$157,890** vs. Female: **\$75,191** - a 2.1× gap



Subscription Underperformance

Subscribers spend **less** (\$59.49 vs. \$59.87); only 27% subscribed;
 $P(\text{Subscriber} \mid \text{Repeat Buyer}) = 0.074$

Weak Acquisition Funnel

Only **83 New** customers vs. 3,116 Loyal - inverted funnel

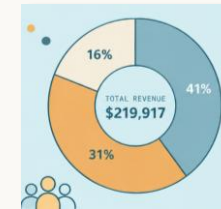


Heavy Discount Dependency

5 core products discounted on ~48% of transactions; **21.5%** of customers are high-spending discount users

Age Concentration Risk

Middle-aged + Adult = **72%** of revenue; Young Adults only 16%



Business Recommendations (1/2)



Rebuild Subscription Value

Target 3,116 Loyal customers with redesigned perks.

Goal: 27% → 40% rate; subscribers out-spend by ~10%.

Impact: +62% subscription revenue (~\$101,400)



Female Acquisition Push

Address 2.1× gender revenue gap via targeted campaigns and category expansion.

Goal: Close gap to 1.5×.

Impact: +\$30K incremental revenue



Fix Acquisition Funnel

Only 83 New customers — invest in referrals and first-purchase discounts.

Goal: 2% → 10% of base.

Impact: 15–20% better 12-month retention

Business Recommendations (2/2)



Tighten Discount Policy

Reduce blanket discounts on Hat, Sneakers, Coat, Sweater, Pants from ~48% → ~25% of transactions.

Goal: Target price-sensitive segments only.

Impact: Recover **8–12% margin** on these 5 lines



Young Adult Strategy

Launch social-first marketing, express shipping promos, and entry-level pricing.

Goal: Grow revenue share from 16% → 22%.

Impact: +\$32,000 incremental revenue

Thank You

Questions, suggestions, or ideas?

Let's connect and explore opportunities for future project collaborations.

✉ | [LinkedIn](#) | [GitHub](#)

